

Electronic Circuits

Prof. Nizamettin AYDIN

naydin@yildiz.edu.tr

<http://www.yildiz.edu.tr/~naydin>

Dr. Gökhan Bilgin

gokhanb@ce.yildiz.edu.tr

1

FET Amplifiers

FETs provide:

- Excellent voltage gain
- High input impedance
- Low-power consumption
- Good frequency range

2

FET Small-Signal Model

Transconductance

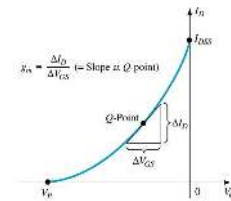
The relationship of a change in I_D to the corresponding change in V_{GS} is called **transconductance**

Transconductance is denoted g_m and given by:

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}}$$

3

Graphical Determination of g_m



4

Mathematical Definitions of g_m

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}}$$

$$g_m = \frac{2I_{DSS}}{|V_P|} \left[1 - \frac{V_{GS}}{V_P} \right]$$

Where $V_{GS} = 0V$ $g_{m0} = \frac{2I_{DSS}}{|V_P|}$

$$g_m = g_{m0} \left[1 - \frac{V_{GS}}{V_P} \right]$$

Where $1 - \frac{V_{GS}}{V_P} = \sqrt{\frac{I_D}{I_{DSS}}}$

$$g_m = g_{m0} \left(1 - \frac{V_{GS}}{V_P} \right) = g_{m0} \sqrt{\frac{I_D}{I_{DSS}}}$$

5

FET Impedance

Input impedance:

$$Z_i = \infty \Omega$$

Output Impedance:

$$Z_o = r_d = \frac{1}{y_{os}}$$

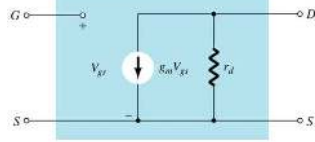
where:

$$r_d = \frac{\Delta V_{DS}}{\Delta I_D} \Big|_{V_{GS} = \text{constant}}$$

y_{os} = admittance parameter listed on FET specification sheets.

6

FET AC Equivalent Circuit

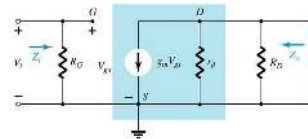
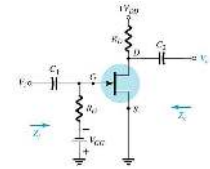


7

Common-Source (CS) Fixed-Bias Circuit

The input is on the gate and the output is on the drain

There is a 180° phase shift between input and output



8

Calculations

Input impedance:

$$Z_i = R_G$$

Output impedance:

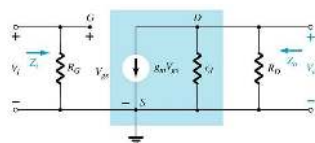
$$Z_o = R_D \parallel r_d$$

$$Z_o \approx R_D \quad | \quad r_d \geq 10R_D$$

Voltage gain:

$$A_v = \frac{V_o}{V_i} = -g_m (r_d \parallel R_D)$$

$$A_v = \frac{V_o}{V_i} = -g_m R_D \quad | \quad r_d \geq 10R_D$$

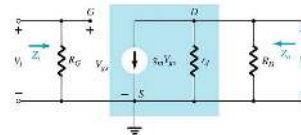
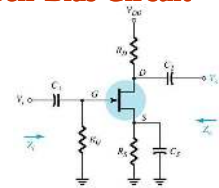


9

Common-Source (CS) Self-Bias Circuit

This is a common-source amplifier configuration, so the input is on the gate and the output is on the drain

There is a 180° phase shift between input and output



10

Calculations

Input impedance:

$$Z_i = R_G$$

Output impedance:

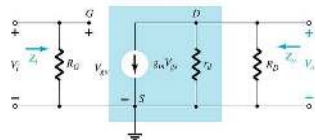
$$Z_o = r_d \parallel R_D$$

$$Z_o \approx R_D \quad | \quad r_d \geq 10R_D$$

Voltage gain:

$$A_v = -g_m (r_d \parallel R_D)$$

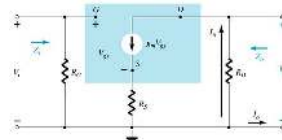
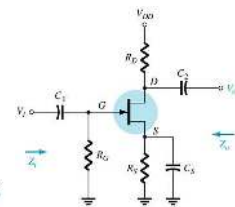
$$A_v = -g_m R_D \quad | \quad r_d \geq 10R_D$$



11

Common-Source (CS) Self-Bias Circuit

Removing C_S affects the gain of the circuit.



12

Calculations

Input impedance:

$$Z_i = R_G$$

Output impedance:

$$Z_o \cong R_D \quad \left| \quad r_d \geq 10R_D \right.$$

Voltage gain:

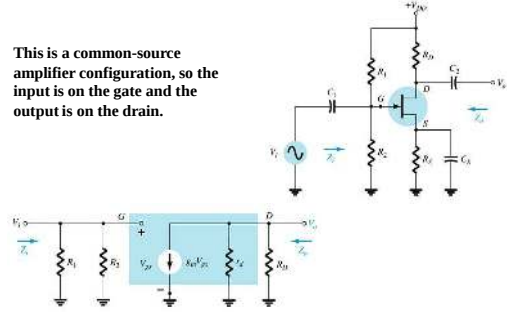
$$A_v = \frac{V_o}{V_i} = -\frac{g_m R_D}{1 + g_m R_S + \frac{R_D + R_S}{r_d}}$$

$$A_v = \frac{V_o}{V_i} = -\frac{g_m R_D}{1 + g_m R_S} \quad \left| \quad r_d \geq 10(R_D + R_S) \right.$$

13

Common-Source (CS) Voltage-Divider Bias

This is a common-source amplifier configuration, so the input is on the gate and the output is on the drain.



14

Impedances

Input impedance:

$$Z_i = R_1 \parallel R_2$$

Output impedance:

$$Z_o = r_d \parallel R_D$$

$$Z_o \cong R_D \quad \left| \quad r_d \geq 10R_D \right.$$

Voltage gain:

$$A_v = -g_m (r_d \parallel R_D)$$

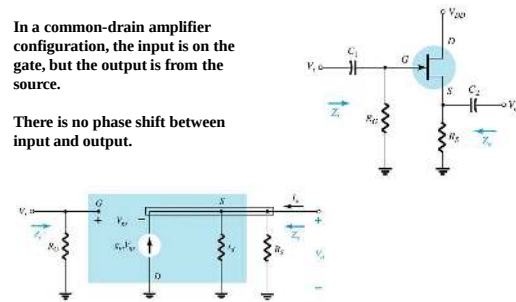
$$A_v = -g_m R_D \quad \left| \quad r_d \geq 10R_D \right.$$

15

Source Follower (Common-Drain) Circuit

In a common-drain amplifier configuration, the input is on the gate, but the output is from the source.

There is no phase shift between input and output.



16

Impedances

Input impedance:

$$Z_i = R_G$$

Output impedance:

$$Z_o = r_d \parallel R_S \parallel \frac{1}{g_m}$$

$$Z_o \cong R_S \parallel \frac{1}{g_m} \quad \left| \quad r_d \geq 10R_S \right.$$

Voltage gain:

$$A_v = \frac{V_o}{V_i} = \frac{g_m (r_d \parallel R_S)}{1 + g_m (r_d \parallel R_S)}$$

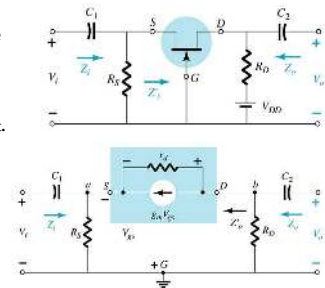
$$A_v = \frac{V_o}{V_i} = \frac{g_m R_S}{1 + g_m R_S} \quad \left| \quad r_d \geq 10 \right.$$

17

Common-Gate (CG) Circuit

The input is on the source and the output is on the drain.

There is no phase shift between input and output.



18

Calculations

Input impedance:

$$Z_i = R_S \parallel \left[\frac{r_d + R_D}{1 + g_m r_d} \right]$$

$$Z_i \approx R_S \parallel \frac{1}{g_m} \quad | r_d \gg 10R_D$$

Output impedance:

$$Z_o = R_D \parallel r_d$$

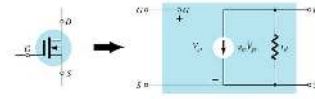
$$Z_o \approx R_D \quad | r_d \gg 10$$

Voltage gain:

$$A_v = \frac{V_o}{V_i} = \frac{[g_m R_D + \frac{R_D}{r_d}]}{[1 + \frac{R_D}{r_d}]} \quad A_v \approx g_m R_D \quad | r_d \gg 10R_D$$

19

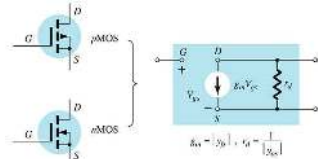
D-Type MOSFET AC Equivalent



20

E-Type MOSFET AC Equivalent

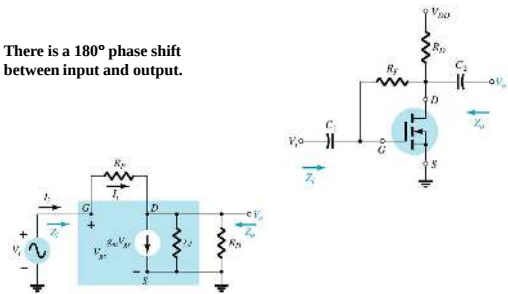
g_m and r_d can be found in the specification sheet for the FET.



21

Common-Source Drain-Feedback

There is a 180° phase shift between input and output.



22

Calculations

Input impedance:

$$Z_i = \frac{R_F + r_d \parallel R_D}{1 + g_m (r_d \parallel R_D)}$$

$$Z_i \approx \frac{R_F}{1 + g_m R_D} \quad | R_F \gg r_d \parallel R_D, r_d \gg 10R_D$$

Output impedance:

$$Z_o = R_F \parallel r_d \parallel R_D$$

$$Z_o \approx R_D \quad | R_F \gg r_d \parallel R_D, r_d \gg 10R_D$$

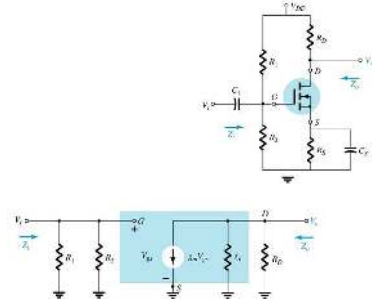
Voltage gain:

$$A_v = -g_m (R_F \parallel r_d \parallel R_D)$$

$$A_v \approx -g_m R_D \quad | R_F \gg r_d \parallel R_D, r_d \gg 10R_D$$

23

Common-Source Voltage-Divider Bias



24

Calculations

Input impedance:

$$Z_i = R_1 \parallel R_2$$

Output impedance:

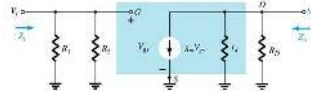
$$Z_o = r_d \parallel R_D$$

$$Z_o \approx R_D \mid r_d \geq 10$$

Voltage gain:

$$A_v = -g_m(r_d \parallel R_D)$$

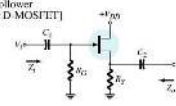
$$A_v \approx -g_m R_D \mid r_d \geq 10 R_D$$



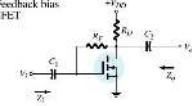
25

Summary Table

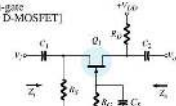
Source-follower
[JFET or D-MOSFET]



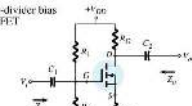
Drain Feedback bias
E-MOSFET



Common-gate
[JFET or D-MOSFET]



Voltage-divider bias
E-MOSFET

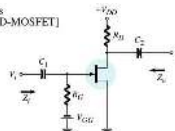


more...

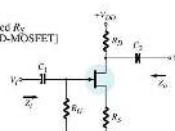
26

Summary Table

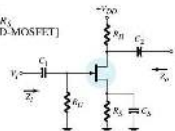
Fixed-bias
[JFET or D-MOSFET]



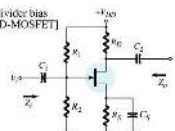
Self-bias
Unbypassed R_S
[JFET or D-MOSFET]



Self-bias
bypassed R_S
[JFET or D-MOSFET]



Voltage-divider bias
[JFET or D-MOSFET]



27

Troubleshooting

Check the DC bias voltages:

If not correct check power supply, resistors, FET. Also check to ensure that the coupling capacitor between amplifier stages is OK.

Check the AC voltages:

If not correct check FET, capacitors and the loading effect of the next stage

28

Practical Applications

Three-Channel Audio Mixer

Silent Switching

Phase Shift Networks

Motion Detection System

29